QEP (MPOS) User Guide

Power Application Controller^{TM}

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1 DESCRIPTION

The QEP software module is one of the selectable MPOS (Motor Position) options available to the user for motor control using the Active-Semi FOC firmware. When selected, the QEP peripheral available in the MCU is configured to decode the signals from a QEP encoder fitted to a motor assembly.

This method of sensing the rotor position is typically used in applications where:

1) A high level of rotor angle accuracy is necessary
2) The motor must operate at speeds below the capability of sensor-less motor position methods

The QEP implementation is currently only available in combination with the Align&Go startup method.

Note that QEP is only supported by PAC55xx products.

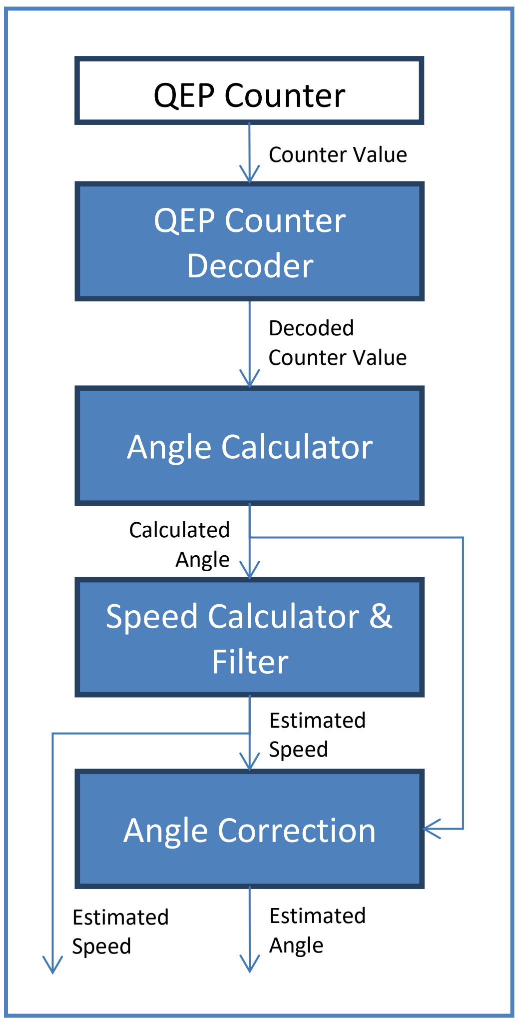
2 FIRMWARE IMPLEMENTATION

2.1 QEP Process

The FOC v4.1 firmware QEP mode processes the information from the QEP hardware peripheral to calculate the exact motor position and speed as outlined in the following steps:

1) The QEP peripheral uses the quadrature encoded signals from the motor's QEP encoder to generate a counter value that represents the motor position.
2) The QEP processing block decodes the counter value to eliminate counter overflow and also takes into consideration the selected motor direction.
3) The mechanical rotor angle is calculated from the decoded QEP counter value and the electrical angle is derived.
4) Rotor speed is calculated from the change in angle each control period and a low-pass filter is applied.
5) Angle correction is applied with a constant offset component.

Below is a block diagram that describes how the output from the QEP block is processed.



2.2 Decoding the QEP counter

The QEP counter must be decoded before it can be converted into a valid mechanical angle value. The reason for this is that when the QEP counter is counting in the negative direction (decreasing value), when the counter reaches 0, it overflows into the maximum counter 16-bit value of 65535 instead of to the maximum QEP resolution. This value must then be re-mapped so that values that overflow into 65535 can be interpreted as the maximum possible counter value according to the QEP encoder resolution.

The following hypothetical example is illustrated in *Figure 1*. Take for example a QEP encoder with a resolution of 8192. In quadrature mode the maximum value the counter will see when moving in the forward direction is 8192 * 4 = 32768 before the index pulse resets the counter to 0. Therefore if the user has defined forward as the desired direction, and the motor is moving in this forward direction, then the counter does not need to be manipulated. But suppose that the counter value is 10000 and at this point the load starts moving the motor in the reverse direction, then the counter value starts counting down. When it reaches 0 it overflows up to 65535 because it does not know that the maximum counter resolution is 32768. At this point, the values need to be remapped so that overflow values are shifted down to be within the range of [0, 32768].

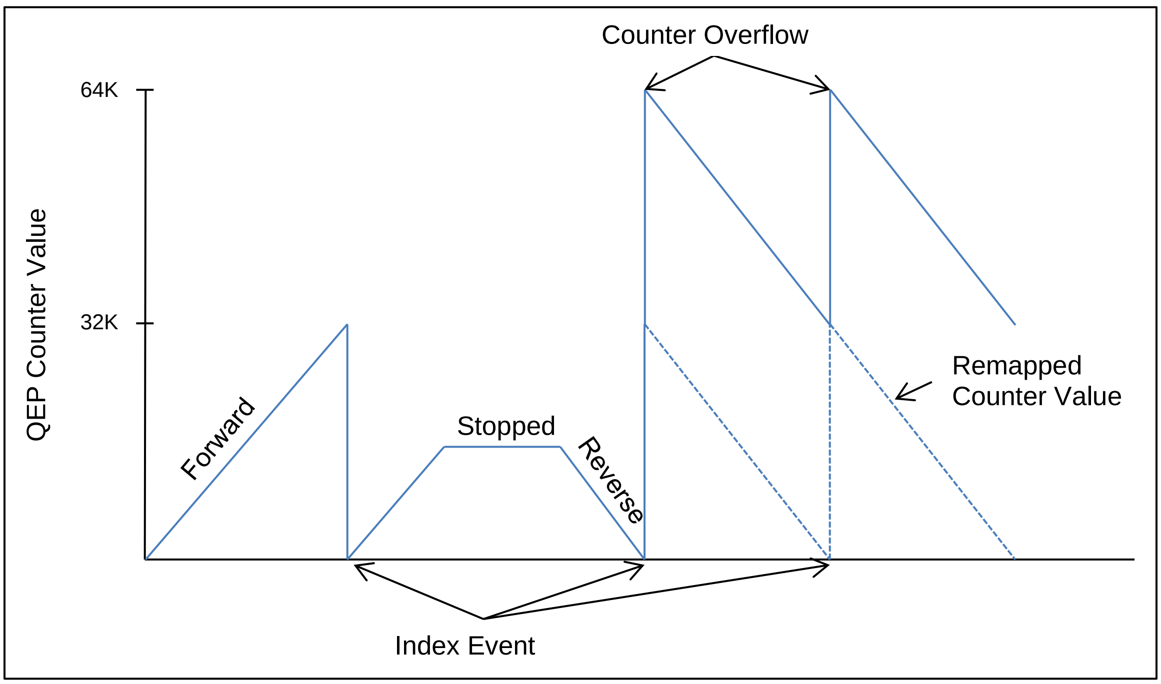


Figure 1: Example of QEP Counter Decoding

2.3 Startup Sequence

When first starting the motor, the QEP block does not know the actual position of the motor until the first index pulse is received. Therefore when starting the motor a startup method like Align&Go is necessary. Once the first index pulse is detected then the QEP peripheral counter will start to correctly represent the mechanical angle of the rotor and the firmware can switch from the Align&Go forced angle to the QEP angle.

Figure 2 depicts the sequence followed during the motor startup process from zero-speed to QEP mode.

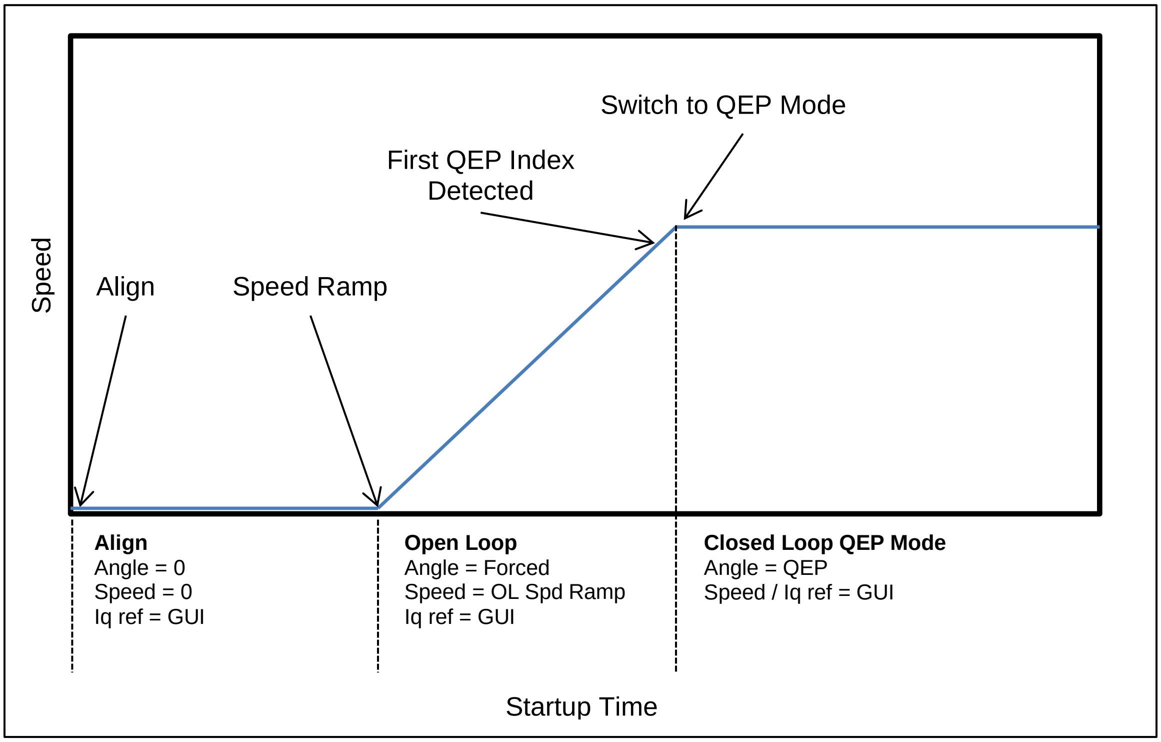


Figure 2: Motor Startup Sequence in QEP Mode

2.4 Inputs

The QEP process block receives the following inputs during runtime:

Input Variable Name	Format	Units
counter_counts	Q0	-

2.5 Outputs

The QEP process block produces the following outputs:

Output Variable Name	Format	Units
qep_angle	Q16 (1.0 = 65536)	rad [$-\pi$, π]
qep_spd_radps	Q16 (1.0 = 65536)	rad/sec
qep_spd_scaled_q14	Q14 (1.0 = 16384)	Normalized speed [-1, 1]
qep_spd_scaled_q28	Q28 (1.0 = 268435456)	Normalized speed [-1, 1]

2.6 Faults

2.6.1 QEP Speed Error Warning

During the Align&Go startup, the QEP speed is compared with the forced speed. If this feature is enabled, and if the error is larger than the threshold, the firmware will not switch over to QEP mode and will remain in the Align&Go mode. The main status window notifies warns the user if this occurs.

If this feature is not enabled, the firmware will do a blind switch-over from Align&Go to QEP mode.

2.6.2 QEP Speed Error Timeout

If this feature is enabled, and if the QEP speed does not converge with the Align&Go speed within the specified number of mechanical revolutions, then the motor will be disabled and the fault is displayed in the main status window.

If this feature is not enabled, the firmware will remain in Align&Go mode until the motor is disabled.

3 Hardware Configuration

3.1.1 QEP Timer Block

The main steps in configuring the QEP Timer block are summarized below:

1) Configure timer clock source and divider
2) Configure block to run in quadrature mode or in single edge mode
3) Configure QEP counter to get reset to 0 on index event
4) Conficure QEP interrupts to trigger only on index event
5) Enable the QEP timer
6) Enable NVIC interrupt for QEP timer and configure priority

The QEP peripheral block is configured to generate an interrupt only to detect the first index pulse after startup. This interrupt indicates to the startup controller that the QEP angle is ready to be used and that the MPOS module can start using the QEP angle instead of the Align&Go forced angle.

After the first QEP index interrupt is triggered, this interrupt is disabled and QEP interrupts are no longer used.

The QEP timer configuration is done in “qep_timer_config.h”.

3.1.2 QEP Port

QEP operation requires the use of 3 port inputs:

- 2 inputs for quadrature signals A and B
- 1 input for index signal

All 3 ports inputs used for the QEP block are configured as follows:

1) Configure port peripheral MUX to select QEP peripheral. This configuration depends on which ports are available on a particular device. The PCB design must take into account the ports that are available for the QEP peripheral.
2) Configure the 3 QEP port pins as inputs.
3) Disable internal pullups for the 3 QEP port pins.
4) Disable internal pulldowns for the 3 QEP port pins.

The QEP port configuration is done in “qep_io_config.h”

4 SETTING UP QEP

4.1 Selecting QEP Module

The QEP module is enabled in “config_app.h” by selecting the following definition:

```
#define MPOS_SELECT                    MPOS_QEP_START_ALIGNGO
```

Selecting this MPOS option will automatically also enable the Align&Go module for startup.

4.2 Compile Options

The following compile options are available in “config_app.h” to configure the QEP process block:

`ENABLE_ANGLE_CORRECTION`

Apply angle correction to calculated angle consisting of a constant offset component.

Recommended Value: ENABLED

`ENABLE_QEP_QUAD_MODE`

Run QEP peripheral in quadrature mode.

Recommended Value: ENABLED

`ENABLE_OLSPEED_ANGLEDIFF_CHECK`

Check that the QEP speed error compared to Align&Go speed is under a certain configurable threshold. If check fails, a warning is displayed.

Recommended Value: ENABLED

`ANGLEDIFF_CHECK_TIME_CNTS`

Systick timer divider for checking speed

Recommended Value: 1

`ANGLEDIFF_SPEED_ERROR_PCT_Q16`

Speed error threshold entered as the decimal equivalent of the desired percentage of Align&Go speed setting

Recommended Value: 0.10 (10%)

`ENABLE_OLSPEED_ANGLEDIFF_TIMEOUT`

Enables timeout for QEP speed check. If timeout is exceeded, the motor is disabled.

Recommended Value: ENABLED

`NUM_QEP_MECH_ROTATION`

Speed check timeout in QEP mechanical rotations

Recommended Value: 3

4.3 GUI Configuration

The GUI contains a section where the user can configure the parameters for QEP operation which can vary by motor. The QEP tuning section is shown in *Figure* 3.

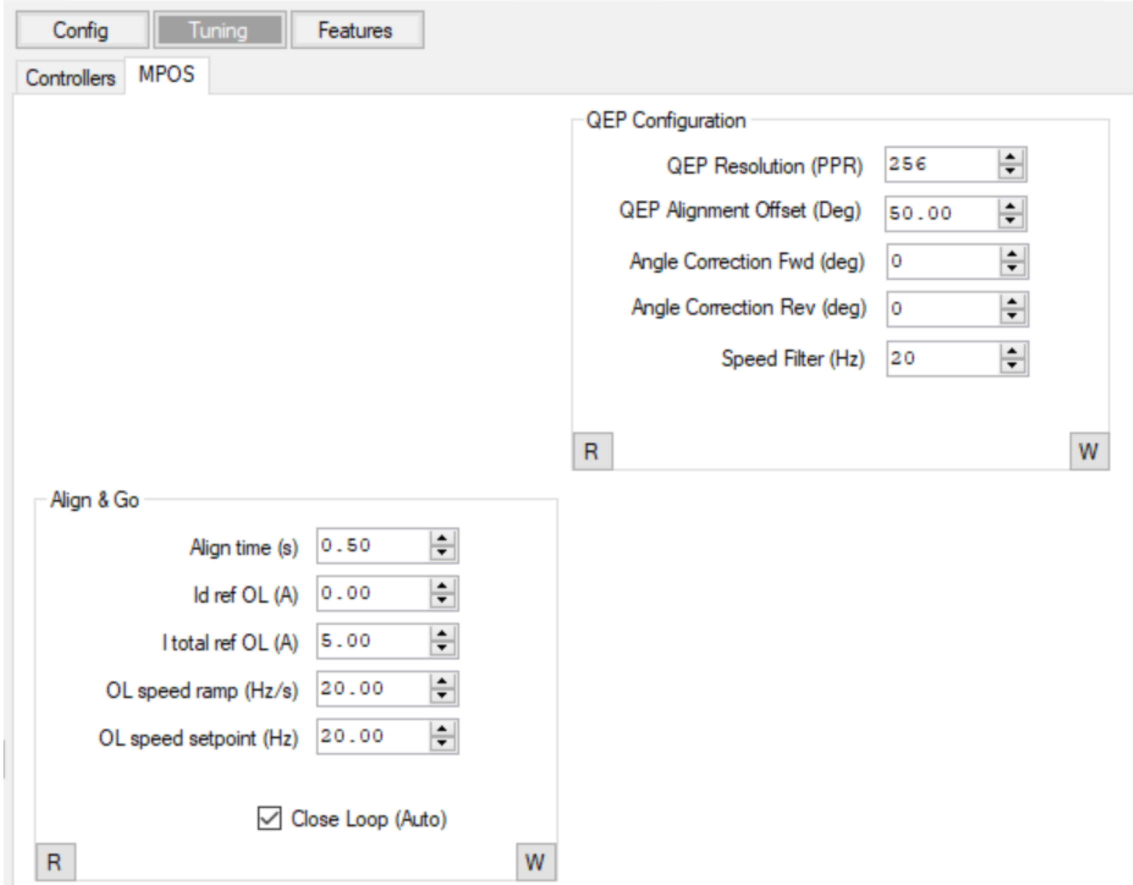


Figure 3: QEP Tuning Section in GUI

4.3.1 QEP Configuration

The QEP configuration section in the GUI can be found by selecting the “Tuning” view and then the “QEP Configuration” tab requires the user to configure the following items:

QEP Resolution (PPR)

The resolution of the QEP encoder is displayed on GUI and it can be defined in the firmware by #define QEP_PULSE_PER_RESOLUTION.

This is the base resolution of the encoder when in single-edge detection mode. Typically, the resolution of a QEP encoder is listed in the specifications as single-edged, which means that in quadrature mode the resolution is 4x higher. Regardless of whether the QEP is operated in single-edge or quadrature mode, the value displayed in this box is the single-edged resolution. The maximum resolution supported is 8192.

QEP Alignment Offset (Deg)

This is the mechanical offset between the motor phase U and the index signal. This offset angle tuning can be found from Section 4.3.2.

Angle Offset Fwd (deg)

This is the offset applied to the QEP angle in the forward direction. This value depends on the mechanical alignment of the QEP index pulse relative to the electrical angle of the rotor. Currently there is no tool to assist the user in tuning this value, so the user will need to obtain this value by trial and error.

Angle Offset Rev (deg)

Same as “Angle Offset Fwd” above but for the reverse direction.

Speed Filter (Hz)

Enter the desired speed filter cutoff frequency. A value of 75Hz should be a good starting point for most applications.

4.3.2 QEP Alignment Offset tuning

Once the QEP sensor is mounted on the shaft, it is required to extract the QEP alignment offset angle while detecting QEP Index signal. This tuning feature intends to automatically calculate the initial offset angle. Here are the steps to use this feature as shown in Figure 4:

a. Enable “#define **ENABLE_QEP_OFFSET_TUNING**” from “config_features.h”.
b. Run the motor in **Open loop mode without Load**.
c. Set "Itotal ref OL" with an initial value (Be sure that Motor can be rotated with this value) and "Id ref OL" = 0.
d. Set "OL speed setpoint" as a slow speed (i.e., 2 Hz or so).
e. Run the motor and check if the motor is rotating smoothly (Current reference could be increased if not smoothly rotating).
f. Set "Id ref OL" equal to "Itotal ref OL" while Motor is running.
g. Start the Offset calculation from GUI as shown in Figure 4.
h. The QEP offset angle is shown in GUI, which is updated once detecting each QEP index.
i. Please copy QEP offset angle to “PUSER_VALUE_QEP_ALIGNOFFSET_DEG” or configure “QEP Alignment Offset” in GUI for QEP regular operation.

NOTES: This offset angle may be affected by the motor cogging torque! It is recommended to do the offset angle fine tuning based on this result!

Please disable “ENABLE_QEP_OFFSET_TUNING” after tuning done because QEP configuration needs to be restored for regular QEP operation.

QEP resolution has been predefined in the firmware side. The GUI box of “QEP Resolution (PPR)” is for display only. If it is shown as 0, please click the button “R” on the corner.

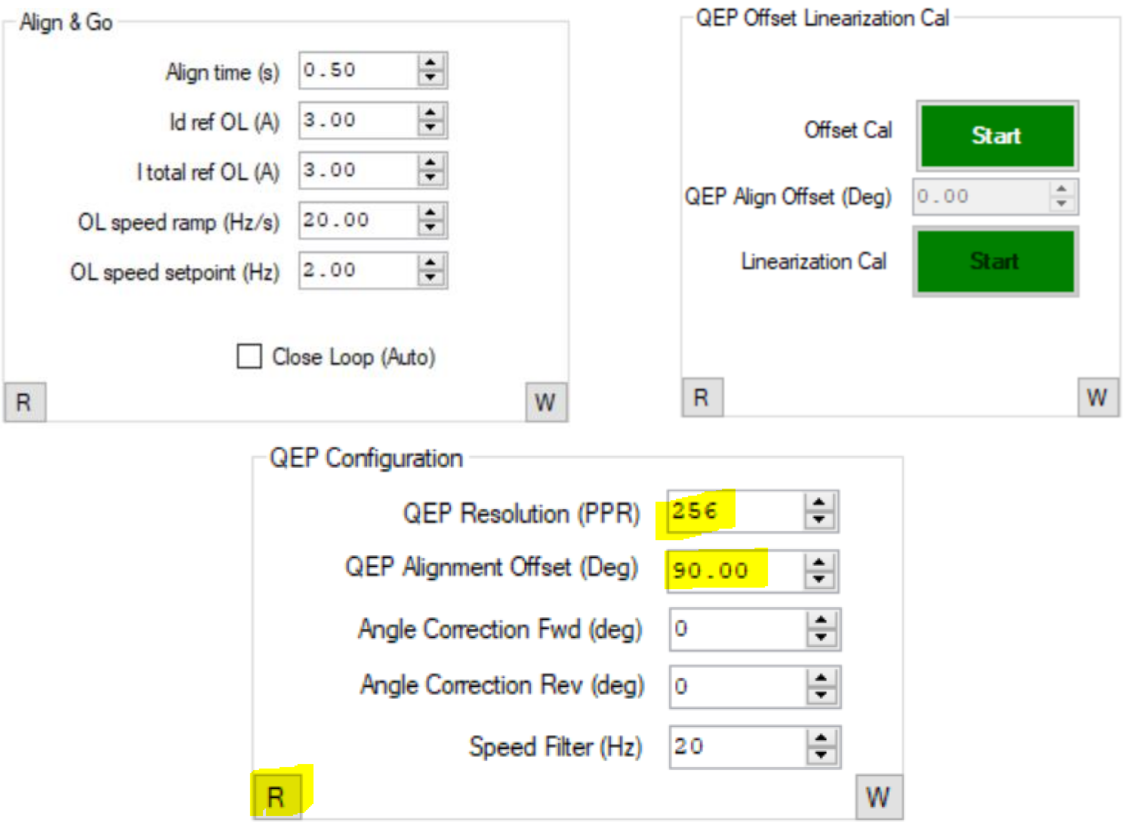


Figure 4: QEP Offset configuration in GUI

4.3.3 Align&Go Startup

The Align&Go section in the GUI requires the user to configure the following items:

Align Time (s)

Enter the align time that is appropriate for the system. Generally the higher the inertia of the motor load, the longer the align time needs to be. A good starting point may be 1s and it can be decreased or increased as necessary to ensure a reliable startup in the shortest possible time.

Id ref OL (A)

For QEP startup this value is not relevant and should always be set to 0.

I total ref (A)

Total starting current for the align&go startup. Increase this value to make sure that enough torque is applied to the rotor and ensure a reliable startup. A good starting point may be 20% of maximum current scale. It may be higher or lower depending on the motor's cogging torque properties and on the load.

OL speed ramp (Hz/s)

Enter the desired speed ramp rate during the align&go startup. The higher the ramp rate the faster the startup will be, but also may require higher value for "I total ref".

OL speed setpoint (Hz)

Enter the final speed target for the align&go startup. It is normal for this final target speed to not be reached since the firmware will end the align&go startup as soon as it detects the first index pulse event and switch to QEP mode.

Close Loop (Auto) Checkbox

This box should be checked always for automatically switching over from align&go mode to QEP mode.

CONTACT INFORMATION

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